



Proposal of a Master's Degree Project in Computer Science



Title

Smart AI Agent for Personalized Message Generation and Campaign Automation in Ooredoo Telecom Marketing

General Context

The main objective of this project is to create an end-to-end AI solution that handles the whole process of customer messages creation and personalization in a telecom environment. In this regard, the system to be developed should send out notifications to the customers according to their data such as personality traits, how they use the service, what subscription they have, and how much contact they have had with the company in the past, among others. Therefore, it will be necessary for the system to generate messages that are tailored to each customer segment and that are individual and human-like.

Moreover, the project aims to develop a web-based platform to facilitate campaign management, instant feedback, and validation of AI-created messages by humans. The human reviewers will be checking the messages before they are sent out, so that only those that are right, appropriate, and consistent with the company's marketing goals will be allowed to pass. The AI agent will continue to learn from the incoming data and gradually improve its message generation skill.

The proposed solution will take over most of the telco marketing communication through automation, thus, reducing manual effort and attracting new customers via better engagement strategies. The combination of AI with a real-time operational system not only allows for fast messaging but also maintains the personalized experience that modern customers are looking for. In the long run, the project aims to not only reduce telecom marketing costs and labor through campaign management and message automation but also guarantee large-scale personalization.

Keywords: Natural Language Generation (NLG), Customer Segmentation, Personalization, Telecom Marketing Automation, Campaign Management.

Problem Statement and Objectives

Telecom companies not only have the problem of engaging customers efficiently but also the problem of managing their huge communication campaigns at the same time. The traditional approach to the personalization of messages is usually very slow, error-prone, and lacking in the required flexibility to keep up with changing customer preferences and habits. With this project, Ooredoo company would be able to automate message generation and campaign management through AI and thus communicate with customers at a large scale while ensuring that every single message is personalized and appropriate.

The main obstacle is creating an AI system that would be able to produce not only messages with appropriate context but also change over time in accordance with customer profiles. Additionally, to maintain the high quality of AI content, there should be a validation process that would combine automation with human supervision.

Here are some of the project's specific objectives:

- Develop a machine learning model to analyze and divide customer data making use of their behavioral patterns to produce customized messages by means of natural language generation (NLG).
- Introduce a validation layer consisting of humans (human-in-loop) that would require messages produced by AI to be checked for accuracy and relevance before they are sent out.
- Provide a simple and easy-to-use platform that will let marketing teams work with client data, verify AI-generated messages, and track the results of their campaigns instantly.
- Connecting the telecom systems with the messages produced by AI will allow the clients to receive the approved messages instantly, which in turn will facilitate uninterrupted communications at all customer touchpoints.
- Additionally, the system will be under constant performance assessment to measure customer engagement, and the AI model will be upgraded based on the feedback and performance data.

Work Plan

Phase 1: Literature Review and Requirement Gathering (2 weeks)

- Review literature on the application of machine learning in customer segmentation and NLG in the marketing field.
- Interviewing and discussing the needs and requirements with the staff of the telecom marketing department.

Phase 2: AI Model Development and Data Collection (1 month)

- Collect and preprocess customer data (usage, subscription, and engagement history).
- Train the AI model to classify customers and generate personalized messages using NLG.

Phase 3: Web Platform Development (1.5 months)

- Create a web application with user roles (administrator, validator, analyst).

- Incorporate campaign management, message validation, and real-time reporting as integrated features.

Phase 4: Integration and Testing (1 month)

- Integrate the system with the telecom's SMSI for real-time message delivery.
- Conduct unit, integration, and performance testing to ensure smooth operation.

Phase 5: Evaluation and Refinement (1 month)

- Conduct and present feedback from pilot campaigns, and analyze customer engagement metrics.
- Make adjustments to the system in light of feedback received, thereby enhancing the performance of the AI model and the usability of the platform at the same time.

Selected References

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